Behrooz Moosavi Ramezanzadeh

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Research Interests

- **Calculus of Variations:** $\mathscr{A}$-quasiconvexity, lower semicontinuity, relaxation.
- **Partial Differential Equations:** nonlinear elliptic equations, viscosity solutions, regularity theory.

Education

University of Pittsburgh — Pittsburgh, PA, USA

- *M.S. in Mathematics* **Advisor:** Juan J. Manfredi **GPA:** 3.96/4.00 *Expected May 2027*
  Thesis: *Partial Differential Equations and Stochastic Tug-of-War Games*.
  Committee: Juan J. Manfredi, Armin Schikorra, and Piotr Hajłasz.
  Mathematics Ph.D. preliminary examinations:
  Analysis 60/60; Linear Algebra 57.7/60. Both passed on the first attempt (pass mark: 40/60).
- *Ph.D. student in Economics* *Aug. 2023 – present*
  Advisors: Jordan Awan, Arie Beresteanu **GPA:** 3.84/4.00
- *M.A. in Economics* *Aug. 2025*

Sharif University of Technology — Tehran, Iran

- *B.S. in Mathematical Sciences* **GPA:** 3.78/4.00 **Class rank:** 4/80 *Aug. 2016 – Jan. 2022*

Research Experience

Calculus of Variations — Variable-Coefficient Differential Constraints *Spring 2026 – present*

Carnegie Mellon University. Research advisor: Irene Fonseca.

- Study lower semicontinuity and $\mathscr{A}$-quasiconvexity for variational problems under variable-coefficient linear differential constraints.
- Proved sufficiency under Lipschitz coefficients and pointwise constant rank, allowing spatially varying rank, using localization, blow-up, and frozen-operator projections.
- Proved necessity by adapting the Davoli–Fonseca moving-kernel construction, without pseudodifferential operators, using a Lipschitz kernel-projection estimate under a rank condition weaker than uniform constant rank.

M.S. Thesis — Nonlinear Elliptic PDE and Stochastic Games *Fall 2025 – present*

University of Pittsburgh. Advisor: Juan J. Manfredi.

- Identified the regularized p-Laplacian with the ordinary p-Laplacian of an affine lift one dimension higher; explicitly computed the projected noise kernel and tilted extremal terms of the induced scheme.
- Established existence, uniqueness, and Borel measurability for the projected dynamic programming principle with strip boundary data, using monotone iteration and a comparison principle adapted to the tilted terms.
- Identified the unique solution with the value of an explicit projected tug-of-war game on an augmented state space, using super/submartingale arguments and measurable almost-optimal strategies.
- Proved convergence to the viscosity solution as $\varepsilon \to 0$ for $p \geq 2$ and a pointwise asymptotic mean value characterization for C^2 solutions throughout $1 < p < \infty$; constructed a counterexample at the formal endpoint $p = \infty$.

Publications and Preprints

Mathematics

A Projected Tug-of-War Game for the Regularized p-Laplacian *2026*

Journal of Elliptic and Parabolic Equations. [DOI] [arXiv:2605.01218]

An Asymptotic Mean Value Characterization for the Regularized p-Laplacian *2026*

Preprint. [arXiv:2607.02910]

Optimal Control of an Epidemic with Intervention Design *2025*

Preprint. [arXiv:2503.22928]

Extended Set Difference: An Inverse Operation of Minkowski Summation *2025*

With Arie Beresteanu. Preprint. [arXiv:2412.19779]

Econometrics and Statistics

Partial Identification with Auxiliary Moment Restrictions *2026*

With Arie Beresteanu. Preprint. [arXiv:2607.21807]

Public Good Provision under Locally Private Signals *2026*

With Jordan Awan. Preprint. [arXiv:2606.24013]

Bounds for Restricted Selections of Random Sets *2025*

With Arie Beresteanu. Preprint. [arXiv:2512.04539]

Honors and Awards

Gold Medal, Iranian National Mathematical Olympiad *2014*

National competition, Iran.

Distinguished Student, Sharif University of Technology *2021*

Ranked 4th in a graduating class of 80, Department of Mathematical Sciences.

Iranian National University Entrance Examination *2016*

Ranked 48th in the Mathematics and Physics track.

Talks, Conferences, and Summer Schools

Working Group on the De Giorgi Conjecture *Fall 2026*

Participant, Center for Nonlinear Analysis, Carnegie Mellon University. Organizer: Robin Neumayer.

Frontiers of Mathematical and Computational Aspects of Materials Science *October 2026*

Conference participant, Carnegie Mellon University.

Contributed talk, CFE–CMStatistics 2026 *December 2026*

HTW Berlin, Germany. *Asymptotically Optimal Test Statistics for Simulation-Based Finite-Sample Inference*.

Contributed talks, Midwest Econometrics Group Annual Meeting *October 2025, October 2026*

University of Cincinnati (2026). [Slides 2026] University of Illinois Urbana–Champaign (2025). [Slides 2025]

Topics at the Interface of Partial Differential Equations and Probability *June 2026*

Summer school participant, Department of Mathematical Sciences, Carnegie Mellon University.

Teaching and Research Appointments

University of Pittsburgh *2022 – 2026*

Teaching Assistant: Graduate Stochastic Processes (Industrial Engineering, Fall 2022); Probability & Statistics I (Industrial Engineering, Summer 2023); Intermediate Microeconomics (Economics, Fall 2024); Applied Econometrics I and II (Economics, Spring and Fall 2025); Introduction to Microeconomic Theory (Economics, Spring 2026).

Graduate Research Assistant: Department of Industrial Engineering (Fall 2022 – Spring 2023) — implemented first-order splitting methods (projected gradient, ADMM) and numerical routines for constrained optimization in Julia and Python.

Sharif University of Technology *2017 – 2022*

Instructor: Introduction to Linear Algebra for Engineers (Spring 2019) — sole responsibility for lectures, problem sets, and assessment.

Teaching Assistant: Abstract Algebra I (Spring 2020); Topology I (Spring 2021); Elementary Measure Theory (Spring 2022).

Volunteer Tutor: Yarigaran Center, tutoring child laborers (2017–2018).

Expository Lecture Notes

Causal Inference *2026*

Comprehensive Ph.D.-level lecture notes. [GitHub]

Topology *2021*

Lecture series written as a teaching assistant. [GitHub]

Linear Algebra *2019*

Full lecture series written as course instructor. [GitHub]

An Advanced Introduction to Mathematical Logic and Set Theory *2019*

Expository notes on first-order logic and axiomatic set theory. [GitHub]

Selected Advanced Coursework

Carnegie Mellon Univ.[†]	**Univ. of Pittsburgh**	**Sharif Univ. of Technology**
Calculus of Variations	PDE I	Real Analysis I
Functional Analysis	PDE II[‡]	Real Analysis II
Measure and Integration	Analysis II	Complex Analysis
Advanced Optimal Transport	Analysis III	Topology
Statistical Optimal Transport I	Analysis IV[‡]	Abstract Algebra I
Statistical Optimal Transport II	Differential Geometry I	Dynamical Systems
	Differential Geometry II[‡]	Numerical Analysis
	Mathematics of Finance I, II	Linear Algebra

[†] PCHE cross-registration.
[‡] Enrolled for Spring 2027.

Technical Skills and Languages

- **Scientific computing:** MATLAB, Python, Julia, R, C++.
- **Mathematical software and typesetting:** Lean 4, LaTeX.
- **Software:** Author of `LangevinFlow`, an R package with a C++ backend published on CRAN.
- **Languages:** Persian (native), English (fluent).